

5. (a) What are transducers and how they classified ?
List some advantages of electrical transducer over mechanical transducers. 10
- (b) List the advantage and disadvantage of capacitive transducers. 5
- (c) A capacitive transducer is made up of two concentric cylindrical electrode which are 20 mm in length. The outer diameter of the inner electrode is 3.5mm and a gap of 0.1 mm between the electrodes contain air as the dielectric medium. Find out the change in capacitance when the inner movable electrode is shifted through a distance of 1.5 mm. 5
6. (a) Describe the basic concept of the "Seismic instrument" with a neat sketch. 10
- (b) What are read out devices ? Sketch and explain the working of a cathod-ray-oscilloscope. (CRO) 10
7. Write short notes on the following (Any Four)
- (a) Reed vibrometer
- (b) Bridgman Gauge
- (c) Resistance Strain - gauge
- (d) Optical pyrometer
- (e) Use of strain gauge on rotating shaft. 4x5=20

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Roll No.

B.E. II (M)

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Mech. Meas. Instr.

SECOND B.E. EXAMINATION - 2019

(MECHANICAL ENGINEERING BRANCH)

(THIRD SEMESTER)

ME - 206 A: MECHANICAL MEASUREMENTS

AND INSTRUMENTATION

Time Three Hours

Maximum Marks - 100

Note:- (1) Attempt any FIVE questions.

(2) All questions carry equal marks.

(3) All the parts of a questions should be attempted together.

1. (a) Define the following terms:

Calibration

hysteresis

Threshold.

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(Contd.)

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- (b) The pointer scale of a thermometer has 100 uniform divisions, full scale reading is 200°C and $1/10^{\text{th}}$ of a scale division can be estimated with a fair degree of accuracy. Determine the resolution of the instrument. 4

- (c) During a test run, measurement of temp' were made 100 times with variations in apparatus and procedures. After applying corrections for the known systematic errors, the following data were recorded:

Temp($^{\circ}\text{C}$)	97	98	99	100	101	102	103	104	105
Frequency of Occurrence	2	5	9	19	33	18	8	4	2

Make calculations for the arithmetic mean, the average deviation, standard deviation, variance and the probable error of the reading. 10

2. (a) Describe the principle of operation of linear variable differential transformer (LVDT). Identify the input and output of the system and sketch the typical input and output graph. 10
- (b) State the law of intermediate temp' and the law of intermediate metals. 6
- (c) A platinum resistance temp' detector has a resistance of 100Ω at 0°C , 138.50Ω at 100°C , and

175.83 Ω at 200°C . What will be non-linearity error in 0°C at 100°C if the detector is assumed to have a linear relationship between 0°C & 200°C . 4

3. (a) Explain with a neat sketch the constructional feature and basic working principle of "McLeod Gauge" used for the measurement of low pressure. 10

- (b) In a test on a copper-constantan thermocouple, the hot junction was immersed in an oil bath together with a mercury-in-glass thermometer, and the cold junction was immersed in a melting ice. The temp' of the oil bath was raised from 40°C to 200°C and the thermocouple emf in millivolts was measured at intervals of 40°C . The fine readings of emf thus obtained were as follows:

1.54 3.38 5.37 7.46 9.62

Use this information to plot a calibration curve for the thermocouple, and from it estimate the temp' indicated by an emf of 5.99 mV. 10

4. (a) Why is a rotameter called variable area flow meter? Sketch and explain its working. 10
- (b) State the objective of the flow visualisation. Explain any one method adopted for flow visualisation in low speed flows. 10

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Write short notes on the following (Any Four)

- ~~(a)~~ Types of measurement Errors.
- (b) Bridge man pressure gauge.
- (c) Optical pyrometer.
- ~~(d)~~ Schlieren apparatus
- (e) Use of strain gauges on rotating shafts.
- ~~(f)~~ Analog and Digital instruments. (5+5+5+5)

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B.E. II (M)

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Mech. Meas. Instr.

**SECOND B.E. EXAMINATION-2018
(MECHANICAL ENGINEERING BRANCH)
(THIRD SEMESTER)**

**ME-206A: MECHANICAL
MEASUREMENTS AND
INSTRUMENTATION**

Time - Three Hours
Maximum Marks - 100

Note :- (1) Attempt any FIVE Questions.

- (2) All questions carry equal marks.
- (3) All the parts of a questions should be attempted together.

1. (a) Explain Range and span with example.

(b) A thermometer scale has 100 divisions and full scale reading is 200°C . If $\frac{1}{10}$ division can be read with accuracy, Find resolution.

(c) The discharge Q over a triangular weir is given by the relation $Q=KH^{2.5}$. The uncertainty in the head reading is 2% of the absolute value. What is the % error in the discharge measurement, if no error occurs in other measurements. (4+6+10)

2. (a) What is transducer. Sketch and explain various types of transducers with examples. Discuss their desired characteristics.

(b) List various methods of pressure measurement. Sketch and explain the working of thermal conductivity Gauge. (10+10)

3. (a) Differentiate between Thermister, Resistance temperature Detector and Thermocouple, with their merits and demerits. How reference temperature compensation is done.

(b) A chromel-constantan thermocouple has cold junction at 0°C . The temperature-mV relation is given below.

Temperature $^{\circ}\text{C}$	0	10	20	30	40	50
emf mV	0	.593	1.191	1.8	2.415	3.02

Find the temperature of hot junction if the emf reading is 2.95mV . (10+10)

4. (a) Discuss measurement of temperature by thermoelectric effects. Sketch and explain the working of a radiation pyrometer.

(b) List various methods of flow-measurement. Sketch and explain the working of a magnetic flowmeter. Discuss its merits and demerits. (10+10)

5. (a) A 120 ohm strain gauge having a gauge factor of 2.0 has been mounted on a steel bar and forms one arm of a symmetrical wheat stone bridge circuit. When the steel bar is subjected to a tensile load, the voltmeter indicates an output of 5mV . Find mechanical strain if operating current of strain gauge is 15mA .

(b) Explain various types of strain gauges and gauge factor. Sketch and explain a strain gauge load cell. List various methods of vibration measurement. Sketch and explain the working of a vibration meter and Accelerometer.

(b) What are read out devices. Sketch and explain the working of a (X-Y) recorder. Discuss its merits and demerits.

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Mech.Meas.Instr.

SECOND B.E. EXAMINATION, 2017
(MECHANICAL ENGINEERING BRANCH)
(THIRD SEMESTER)

**ME-206 A: MECHANICAL MEASUREMENTS
AND INSTRUMENTATION**

Time - Three Hours

Maximum Marks : 100

- Note:-
- (1) Attempt any **FIVE** questions.
 - (2) All questions carry equal marks.
 - (3) All the parts of a questions should be attempted together.

1.
 - (a) Differentiate between accuracy and precision which is most desirable in measurement and why? 4
 - (b) Define and explain functions of each element in a Generalized measurement system. Identify various functional element in a Bourdon tube pressure gauge. 8
 - (c) A strain gauge of 100 ohm is required. Two gauges of (200 ± 0.1) ohms and two gauges of (50 ± 0.2) ohms are available. Which should be used.
 - (i) Series arrangement with 50 ohms gauges.

OR

- (ii) Parallel arrangement with 200 ohms gauges?

- Calculate the uncertainty of each arrangement. 8
2. (a) What is meant by a transducer? Classify the transducers based on the electrical principles involved. Give at least two examples for each classification. 10
 - (b) Sketch and explain the working of a hot filament ionization gauge. List its advantages and disadvantages. 10
 3. (a) Sketch and explain the working of a bimetallic thermometer with its applications. 10
 - (b) Explain law of intermediate temperatures and law of intermediate metals used in thermocouples. Discuss the influence of these laws on the industrial uses of thermocouples. 10
 4. (a) Explain the working principles of Hot-wire anemometer with suitable sketch of CTA circuit. 10
 - (b) Sketch and explain the working of a lens type schlieren technique. 10
 5. (a) What are strain gauges? Define gauge factor of a resistance strain gauge. Show that for a wire resistance strain gauge, it is given by:

$$=F - 1 + 2\mu$$

Where μ is the poisson's ratio. Under what conditions does the above relation holds good? 10
 - (b) List various methods of angular velocity measurement. Explain the photocell method with neat sketch. 10
 6. (a) What is the principle of working of LVDT? Explain with the help of neat sketch. Describe its applications. 10
 - (b) Explain with neat sketch the construction and working of Cathode ray oscilloscope. 10